Task 3 — OCR Error and Data Quality Analysis

1. OCR Errors Identified

Document	OCR Errors Identified
C1	Most normal printed text was recognized correctly. Errors were mainly observed in scientific notation, mathematical symbols, variables/subscripts, footnotes, and content associated with diagrams.
C2	Normal printed paragraphs were recognized with high accuracy. Minor errors were mainly observed in mathematical expressions, special symbols, and equation references.
C3	Significant errors were observed in mathematical notation, Greek/special symbols, equations, bra-ket notation, subscripts/superscripts, and matrix-related representations.
K1	Errors were mainly related to numbers, decimal values, table entries, and row-column structure. Some numerical characters and table content were not recognized correctly.
K2	OCR errors occurred mainly in numbers, abbreviations, special characters, columns, and structured form/table information. Table/form structure was not always preserved correctly.
K3	Errors were mainly found in equations, mathematical symbols, technical parameters, subscripts/superscripts, figure labels, and technical terminology.

2. Error Categorization

Document	Handwriting	Low Resolution	Skew	Equations / Symbols	Tables	Diagrams / Figures
C1	No	Minor	No major issue	Yes	No	Yes
C2	No	Minor	No major issue	Minor	No	No significant issue
C3	No	Minor	No major issue	Major	No	Major — matrix/notation
K1	No	Minor	No major issue	No	Major	No
K2	No	Minor	No major issue	No	Major	No
K3	No	Minor	No major issue	Major	Minor	Major — figures/labels

3. Easy / Moderate / Difficult Classification

Document	Difficulty	Reason
C1	**Moderate**	Mostly clean printed text, but scientific notation, mathematical symbols, variables/subscripts, and diagram-related content increase OCR difficulty.
C2	**Easy**	Mostly clean printed paragraphs with only minor errors in mathematical expressions and special symbols.
C3	**Difficult**	Heavy mathematical notation, equations, Greek/special symbols, bra-ket notation, subscripts/superscripts, and matrix representations.

Document	Difficulty	Reason
K1	**Moderate**	Numerical data and table structure create additional OCR challenges even though the content is printed.
K2	**Moderate**	Numbers, abbreviations, special characters, columns, and form/table structure make recognition and layout preservation harder.
K3	**Difficult**	Technical equations, symbols, parameters, subscripts/superscripts, figure labels, and technical terminology create substantial OCR challenges.

4. Final Overall Observations

1. Clean, printed paragraphs are generally recognized with good accuracy.

2. Mathematical and scientific content is more difficult for OCR because symbols, equations, subscripts, superscripts, and special notation are easily confused.

3. Tables create both character-recognition errors and structural errors, especially for numbers and row-column alignment.

4. Diagrams and figures are not reliably represented by plain OCR text, particularly when labels or mathematical notation are present.

5. Handwriting and severe skew were not major error sources in the selected documents.

6. Document complexity has a strong effect on OCR quality: C2 was the easiest, while C3 and K3 were the most difficult.

7. Preprocessing does not necessarily improve every document; its effectiveness depends on the original image quality and the type of content present.

Overall Difficulty Order

Easy	Moderate	Difficult
C2	C1, K1, K2	C3, K3